Following the Information Bound from a Search-Window Prior with Sequential Bayesian Experiment Design in Nitrogen-Vacancy Magnetometry

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[Written last, once the simulation and lab results are in.]

I. INTRODUCTION

Nitrogen-vacancy (N-V) centers in diamond measure magnetic fields at room temperature with nanometer spatial resolution [1, 2], and are now used in settings ranging from condensed-matter physics to biology [3–5].

The most widely used readout is optically detected magnetic resonance (ODMR): the photoluminescence of the center is recorded while a microwave drive is stepped through a search window, the range of drive frequencies in which the spin resonance can lie, and the field is read from the position of the resonance. Every point of this scan takes time, so the number of measurements needed to locate the resonance to a given precision sets both the sensitivity and the bandwidth of the magnetometer. It matters most where the scan is repeated, as when a time-varying field is tracked [6, 7] or imaged pixel by pixel [8, 9].

Locating the resonance is an instance of Hamiltonian learning: the parameters of a quantum system's Hamiltonian are inferred from measurements whose settings, here the microwave drive frequency, are chosen adaptively from what has been learned so far [10, 11]. For N-V centers, Hamiltonian learning and adaptive Bayesian estimation have been demonstrated mainly in the time domain, with Rabi and Ramsey sequences in which an evolution time is the chosen setting [12–15]. In ODMR, sequential Bayesian experiment design (SBED) has been applied directly [16]. SBED maintains a posterior distribution over the parameters, here represented by a sequential Monte Carlo (SMC) particle cloud, and chooses each drive frequency by maximizing an expected utility, such as the expected information gain [10]; it reaches a given precision more than an order of magnitude faster than a sweep, the uniform scan conventionally used [16].

Non-adaptive strategies reduce the number of points instead, by measuring a subset of the window and reconstructing the spectrum from it, with a low-complexity reconstruction [17] or with compressed sensing, which matches the accuracy of a full scan with a small fraction of its points [7]. A further line of work learns the measurement policy in advance rather than computing each setting from the posterior. In quantum metrology in general, learned policies have been used for adaptive phase estimation [18, 19], reinforcement learning has been used to find controls that maximize the sensitivity of a quantum sensor [20–22] and to estimate several parameters simultaneously in a photonic experiment [23], and neural networks have replaced the costly optimization of the expected gain in Bayesian design with a fast learned heuristic [24]. For N-V centers, model-aware reinforcement learning on a particle-filter posterior has been applied to Ramsey magnetometry [25, 26], and for a wide field range a Bayesian neural network that narrows the range has been combined with a reinforcement-learning agent that refines the estimate [27].

These studies show that adaptive and learned strategies locate the resonance with far fewer measurements than a sweep, and measure the gain against that sweep. Information bounds are already used as benchmarks in Hamiltonian learning: the Bayesian Cramér–Rao bound certifies how close an SMC estimator comes to the best precision for the measurements it has chosen [10]. Such a bound has not yet been applied to the ODMR search as a whole, starting from nothing more than the search window. It would give a strategy-independent reference, showing how close current strategies come to the best achievable performance and which of their measurements are unavoidable. It would also supply a mechanism for the speedup over a sweep, including its observed growth as the line occupies a smaller fraction of the window [16].

Here we derive this limit for the ODMR search and use it to analyze SBED and drive-frequency sweeps for continuous-wave (CW) and pulsed ODMR. The limit follows from Bayesian estimation theory with the search window as the prior [28, 29], and the information it counts is the same quantity that SBED maximizes. It covers the whole run: finding the resonance, narrowing in on it and refining the estimate are limits of a single bound rather than separate mechanisms. The same bound gives the speedup over a sweep: for a precise target it is approximately the ratio of the window width to the linewidth, independent of the signal-to-noise ratio, which explains its growth with the window. It also recovers the standard CW-ODMR sensitivity as the refine limit, reached only by a strategy that measures at the steepest slope of the line, and shows that the window limits the bandwidth while the line shape and the signal-to-noise ratio limit the sensitivity.

Section II derives the bound from the physical model, together with the speedup, sensitivity and bandwidth that follow from it. Section III describes SBED, with its particle filter, as a strategy that follows the bound, and the sweeps it is compared with, both evaluated against the bound. Section IV presents the simulations and the experiment, and Sec. V discusses the limits of the approach.

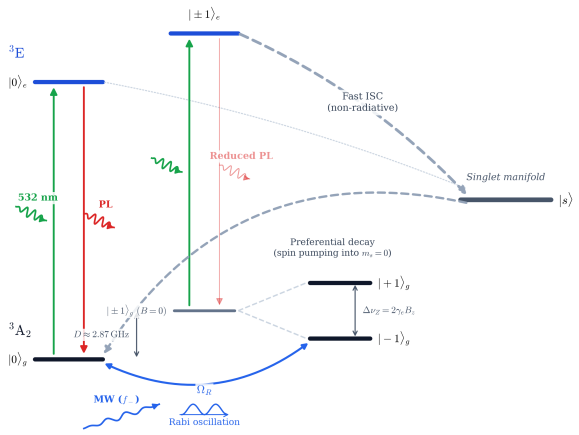


FIG. 1. Five-level model of the N-V center [3, 30]. [Caption to follow the figure.]

II. THEORY

A. Physical model

The negatively charged N-V center is described by five levels [3, 30] (Fig. 1): the spin-triplet ground state $^3A_2$, split into $|0\rangle_g$ and $|\pm1\rangle_g$ by the spin projection m_s on the N-V axis; the spin-triplet excited state ^3E, with $|0\rangle_e$ and $|\pm1\rangle_e$; and a metastable spin singlet $|s\rangle$ that stands for the singlet manifold. Green light drives the spin-conserving optical transitions $|m_s\rangle_g \to |m_s\rangle_e$, which decay back by emitting red photoluminescence (PL). From $|\pm1\rangle_e$, however, the intersystem crossing to the singlet is much faster than from $|0\rangle_e$, and the singlet decays preferentially to $|0\rangle_g$. Two consequences follow. Under continuous illumination the spin is pumped into $m_s = 0$; and a center in $m_s = \pm1$ spends part of its optical cycles in the dark singlet, so it emits less PL than one in $m_s = 0$. The spin state can therefore be prepared and read out optically.

With a static field B_z along the N-V axis, the ground-state spin ($S = 1$) Hamiltonian is [3]

$$H = D\,S_z^2 + \gamma_e B_z S_z + A_\parallel S_z I_z, \tag{1}$$

written in frequency units ($h = 1$), with $D \approx 2.87$ GHz the zero-field splitting, $\gamma_e \approx 28$ GHz/T the electron gyromagnetic ratio, and $A_\parallel$ the axial hyperfine coupling to the nitrogen nuclear spin I ($|A_\parallel| = 2.15$ MHz for ^{14}N, $I = 1$, and 3.03 MHz for ^{15}N, $I = 1/2$ [31]). The strain term $E(S_x^2 - S_y^2)$ is dropped, since it is a small correction when $\gamma_e B_z \gg E$. The eigenstates are the product states $|m_s, m_I\rangle$, and the magnetic-dipole transitions $m_s = 0 \to \pm1$, which conserve m_I, lie at

$$f_\pm(m_I) = D \pm \left(\delta_Z + A_\parallel m_I\right), \qquad \delta_Z = \gamma_e B_z, \tag{2}$$

where the Zeeman shift δ_Z is the distance of each transition from D: the field moves the two transitions apart, symmetrically about D. The hyperfine term splits each of them further into $2I + 1$ lines, one per nuclear projection m_I, spaced by $|A_\parallel|$: a triplet for ^{14}N and a doublet for ^{15}N. Every N-V center contains one of the two isotopes (^{14}N at 99.6% natural abundance, ^{15}N in isotopically enriched or implanted samples), so the hyperfine structure is always present; whether it is resolved depends on the linewidth. We first treat a single line and return to the hyperfine structure below.

An ODMR measurement sets the microwave drive frequency f and records the PL, normalized to its off-resonance level. When f meets a transition, the drive transfers population from $m_s = 0$ to the darker $m_s = \pm1$ state and the PL dips.

The two transitions share the state $|0\rangle_g$ but lie $2\delta_Z$ apart. When this separation is large compared with the linewidth, a drive near f_- is far off resonance with f_+ and couples only $|0\rangle_g$ and $|-1\rangle_g$. The excited state and the singlet then enter only through the rates at which the laser repolarizes the spin and destroys its coherence [32], so the spin reduces to a driven two-level system $\{m_s = 0, m_s = -1\}$. It is therefore enough to model one transition, and we take the one measured in our setup, f_- (Sec. III D); f_+ behaves the same way, mirrored about D. With the zero-field splitting D treated as a known constant, f_- carries the field, $D - f_- = \delta_Z = \gamma_e B_z$, and an uncertainty in f_- is the same uncertainty in the Zeeman shift.

Let the drive have detuning $\Delta = f - f_-$ from the transition and Rabi frequency Ω, the rate in Hz at which the population oscillates on resonance. Off resonance the spin oscillates faster but less deeply, at the generalized Rabi frequency

$$\Omega_{\text{gen}} = \sqrt{\Omega^2 + \Delta^2}.$$

Starting from $m_s = 0$, the probability of finding the spin in $m_s = -1$ after a drive of duration t is

$$P_{-1}(\Delta, t) = \frac{\Omega^2}{\Omega_{\text{gen}}^2}\,\sin^2\!\left(\pi\Omega_{\text{gen}} t\right), \tag{3}$$

and $P_0 = 1 - P_{-1}$ is the probability of finding it still in $m_s = 0$. The prefactor is the depth of the oscillation, the sine its time dependence. Since $m_s = -1$ is the darker state, the PL dip is proportional to P_{-1}. How the drive is applied sets the line shape.

CW ODMR. In continuous-wave ODMR the laser and the microwave drive act on the spin continuously and at the same time, and the PL is read out while the drive is on (Fig. 2). The laser keeps pumping the spin back into $m_s = 0$ while the drive keeps rotating it away, so every optical reset starts a new Rabi oscillation, and the PL, collected over many such cycles, reports Eq. (3) averaged over the time T between resets. When the drive acts for many Rabi periods between resets, $\Omega_{\text{gen}} T \gg 1$, the

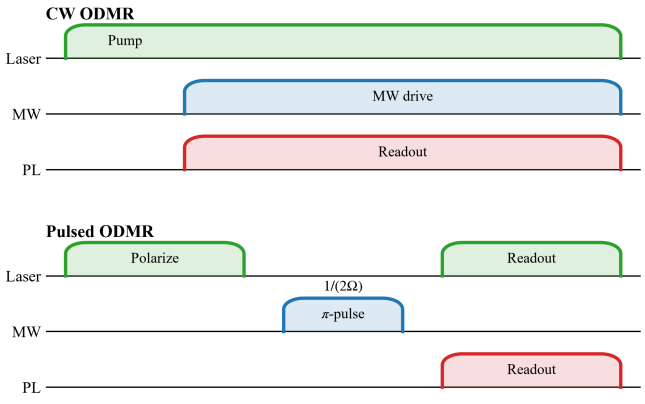


FIG. 2. Laser and microwave sequences of CW and pulsed ODMR. [Caption to follow the figure.]

oscillating factor averages to $\frac{1}{2}$,

$$\bar{P}_{-1}(\Delta) = \frac{\Omega^2}{\Omega_{\text{gen}}^2}\,\frac{1}{T}\int_0^T \sin^2(\pi\Omega_{\text{gen}}t)\,dt \;\longrightarrow\; \frac{1}{2}\,\frac{\Omega^2}{\Omega^2+\Delta^2}. \tag{4}$$

The average only renormalizes the amplitude; the line shape is that of the prefactor, a Lorentzian in the detuning Δ of half width Ω. This is the microwave contribution to the width, power broadening: the stronger the drive, the further off resonance it still flips the spin.

The laser does more than reset the spin. It pumps the spin back toward $m_s = 0$ at a rate Γ_p, and each optical cycle destroys the phase between $m_s = 0$ and $m_s = -1$, at a rate Γ_c; both rates grow with the laser power and saturate at high power. With these rates, and the angular Rabi frequency $\Omega_R = 2\pi\Omega$ of the drive, the line is still a Lorentzian (Appendix A), of half width [32]

$$\gamma = \frac{1}{2\pi}\sqrt{\Gamma_c^2 + \Omega_R^2\,\frac{\Gamma_c}{\Gamma_p}}. \tag{5}$$

The first term is the width the laser alone imposes, through the loss of coherence; the second is power broadening by the microwave, tempered by the pumping. At weak drive the laser sets the width, at strong drive the microwave. Whatever its origin, the line is one Lorentzian dip: the normalized PL expected at drive frequency f is

$$\mu(f) = 1 - C\,\frac{\gamma^2}{\gamma^2 + (f - f_-)^2}, \tag{6}$$

with contrast C, the depth of the dip at $f = f_-$, half width γ of Eq. (5) and full width at half maximum $\Gamma = 2\gamma$. Optical and microwave broadening are not modeled separately: they enter only through the single linewidth Γ, which is inferred with the line.

Pulsed ODMR is a technique that removes the laser from the width of the line [32]. In CW ODMR the laser that reads the spin out also broadens the line through Γ_c, and narrowing it means lowering both powers and with them the contrast and the PL. Pulsed ODMR separates the two in time: a laser pulse polarizes the spin into $m_s = 0$, the laser is switched off while a single microwave π-pulse of duration $t_\pi = 1/(2\Omega)$ is applied, and a second laser pulse reads the spin out (Fig. 2). The pulse fully inverts the spin on resonance, and Eq. (3) at $t = t_\pi$ gives (Appendix A)

$$P_{-1}^{\pi}(\Delta) = \frac{\Omega^2}{\Omega_{\text{gen}}^2}\,\sin^2\!\Big(\frac{\pi}{2}\,\frac{\Omega_{\text{gen}}}{\Omega}\Big). \tag{7}$$

This line is not a Lorentzian. Its half width, $0.80\,\Omega = 0.40/t_\pi$, is the Fourier (time–bandwidth) limit of the pulse: a pulse of duration t_π contains frequencies spread over about $1/t_\pi$, so a longer pulse gives a narrower line. Its side lobes, the first 12% of the depth, are a hazard for a search: a lobe is a dip where there is no resonance. The narrowing stops at $t_\pi \sim T_2^*$, where the spread of the transition frequency from one repetition to the next, of width $\sim 1/T_2^*$, washes out the lobes and sets the width, a width fixed by the spin itself rather than by the laser or microwave power. The laser readout then sets only the depth of the line, the contrast C (Appendix A).

Hyperfine structure. Each nuclear projection m_I gives its own copy of the line, shifted by $A_\parallel m_I$ (Eq. (2)) and weighted by the population of that m_I, so the observed spectrum is a sum of $2I + 1$ lines. Two cases reduce it to the single line of Eq. (6). Near the excited-state level anticrossing the nitrogen nuclear spin is polarized into a single m_I [33]: only one line remains, and its fixed offset $A_\parallel m_I$ is absorbed into f_-. When the line is much wider than the hyperfine spacing, $\Gamma \gg |A_\parallel|$, the lines merge into one.

The second transition. The spectrum is mirror symmetric about D, so the drive window is placed below D and holds the lower transition only. The tail of the upper one, at $f_+ = f_- + 2\delta_Z$, adds $C\gamma^2/[\gamma^2 + (f - f_+)^2]$ inside the window: at the line center a fraction $\gamma^2/(\gamma^2 + 4\delta_Z^2)$ of the contrast C, i.e. 20% at $\delta_Z = \gamma$ and below 1% for $\delta_Z \gtrsim 5\gamma$. Equation (6) assumes $\delta_Z \gg \gamma$. Small Zeeman shifts, where the tails of the two lines overlap, are a separate problem and outside the scope of this work.

f_- is the parameter of interest. The width Γ and the contrast C are nuisance parameters: they are inferred jointly with f_- but only matter through their effect on it. The noise level σ (below) is inferred as well.

Noise. Each reading of the normalized PL is noisy. Its floor is photon shot noise: a reading that collects N_{ph} photons on average off resonance has a relative standard deviation $N_{\text{ph}}^{-1/2}$, and slightly less inside the dip, where fewer photons arrive. Technical noise (laser intensity, microwave power, collection drift) adds to it. We model a single reading, a shot, as

$$y = \mu(f) + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2), \tag{8}$$

with one $\sigma \geq N_{\text{ph}}^{-1/2}$ at every drive frequency; this neglects the small reduction of the shot noise inside the dip,

of relative order of the contrast C. A measurement is the mean of k shots at the same f, so its noise is $\sigma/\sqrt{k}$. The signal-to-noise ratio of a single shot depends on where it is taken: it is the drop in PL at f over the noise,

$$\mathrm{SNR}(f) = \frac{1-\mu(f)}{\sigma} = \frac{C}{\sigma}\,\frac{\gamma^2}{\gamma^2+(f-f_-)^2}. \tag{9}$$

It is largest on resonance, where the drop is the contrast C,

$$\mathrm{SNR} \equiv \mathrm{SNR}(f_-) = C/\sigma, \tag{10}$$

the contrast over the noise, the single number by which a line is labeled. All SNR values in this paper follow Eq. (10). It neglects the mirror line: its tail raises the on-resonance depth to $C[1+\gamma^2/(\gamma^2+4\delta_Z^2)]$, so C/σ is the $\delta_Z \gg \gamma$ limit of the on-resonance SNR.

The prior knowledge of the experimentalist is an interval of width W that holds the resonance, $f_- \in [f_{\min}, f_{\min}+W]$, and the drive frequency is scanned over at least this interval. The ratio W/Γ, the inverse of the fraction of the window the line occupies, sets how hard the search is.

The model above predicts what a measurement records for given parameters. The rest of this section turns the question around: given the measurements, how well can f_- be known, how much can each measurement add, and where should the next one be taken.

B. Information of a measurement

We use the notation of experiment design [34]: the design $x \equiv f$ is the drive frequency at which a measurement is taken, the outcome y is the normalized PL it records, the parameter of interest $\theta \equiv f_-$ is the transition frequency, and $\phi = (\Gamma, C)$ are the nuisance parameters. The expected normalized PL $\mu(x;\theta,\phi)$ of Eq. (6), now written as a function of the design and the parameters, is the mean outcome, and Eq. (8) gives the spread about it.

A measurement at drive frequency x returns one number y, the normalized PL averaged over its k shots. The model predicts the signal $\mu(x;\theta,\phi)$, and by Eq. (8) y scatters about it with variance σ^2/k, the same whatever θ is. The likelihood $p(y \mid x,\theta)$ is the probability of that outcome if the dip is at θ; for Gaussian scatter $\ln p = -k\,(y-\mu)^2/2\sigma^2 + \text{const}$. How much one measurement at x tells about θ is its Fisher information. It asks how strongly an outcome points at θ: take how fast $\ln p$ changes with θ, square it, and average over the outcomes y the measurement could return. The rate of change is

$$\frac{\partial \ln p}{\partial \theta} = \underbrace{\frac{k\,(y-\mu)}{\sigma^2}}_{\substack{\text{how far } y \text{ lies}\\ \text{from the prediction}}} \underbrace{\frac{\partial \mu(x;\theta,\phi)}{\partial \theta}}_{g(x),\ \text{slope}},$$

the slope $g(x)$ being how fast the predicted signal moves when θ moves. Squaring and averaging over y,

$$\begin{aligned} \underbrace{I(x)}_{\substack{\text{information}\\ \text{at } x}} &= \left\langle \left(\frac{\partial \ln p}{\partial \theta}\right)^2 \right\rangle_y \\ &= \left(\frac{k\,g(x)}{\sigma^2}\right)^2 \underbrace{\left\langle (y-\mu)^2 \right\rangle_y}_{\sigma^2/k} \\ &= \frac{\overbrace{k}^{\text{shots}}\ \overbrace{g(x)^2}^{\text{slope}^2}}{\underbrace{\sigma^2}_{\text{shot variance}}}. \end{aligned} \tag{11}$$

The noise enters only through its variance, and the line shape only through the slope g. A shift $\delta\theta$ moves the mean outcome by $g\,\delta\theta$, and it is resolved once that exceeds the noise $\sigma/\sqrt{k}$. The smallest shift a measurement at x resolves is therefore $\delta\theta_{\min}(x) = \sigma/(\sqrt{k}\,|g(x)|)$, and $I(x) = 1/\delta\theta_{\min}(x)^2$. It grows linearly with k, and at the shot-noise limit, $\sigma^2 = 1/N_{\mathrm{ph}}$, with the total number of photons kN_{ph} collected at that drive frequency, which no averaging scheme can beat.

For the Lorentzian of Eq. (6) the slope vanishes on resonance, where the dip is flat, and on the baseline far from it: a measurement at the bottom of the dip or far outside it says almost nothing about where the dip is. The slope is steepest on the two sides of the dip, a detuning $\gamma/\sqrt{3}$ from the resonance, where the dip is 3/4 deep (Appendix B). A measurement there carries the most information one measurement can,

$$I^* = \max_x I(x) = \frac{27}{16}\,k\,\frac{\mathrm{SNR}^2}{\Gamma^2}, \tag{12}$$

with $\mathrm{SNR} = C/\sigma$ of Eq. (10). The best place to measure is thus the flank of the dip, not its center, and the information there grows as $(\mathrm{SNR}/\Gamma)^2$: a deeper or a narrower line is equally valuable. I^* assumes the nuisance parameters (width, contrast) are known; with free nuisances it is an upper bound on the usable information. The nuisances are inferred along with θ but are not the subject of this work.

Measurements are independent given θ, so their Fisher information adds: n measurements at $x_1,\ldots,x_n$ carry

$$J_n = \sum_{i=1}^{n} I(x_i) \;\le\; n\,I^*,$$

with equality only when every measurement is taken at the steepest slope of the true line. By the Cramér–Rao bound [35], an unbiased estimate of θ from them has a standard deviation of at least $J_n^{-1/2} \ge (nI^*)^{-1/2}$.

C. Prior information: the search window

Fisher information measures knowledge in inverse variance, Hz^{-2}: a Gaussian estimate of standard deviation

σ_θ is worth $J = \sigma_\theta^{-2}$, and combining independent Gaussian estimates adds their inverse variances. The same currency measures what is known *before* the first measurement. If the experimentalist knows only that the resonance lies somewhere in the window of width W, the state of knowledge is a uniform prior $p(\theta) = 1/W$ on the window. Its spread is its standard deviation σ_0; with v the distance from the center of the window,

$$\sigma_0^2 = \int_{-W/2}^{W/2} \frac{v^2}{W}\, dv = \frac{W^2}{12}, \qquad J_0 \equiv \sigma_0^{-2} = \frac{12}{W^2}. \quad (13)$$

A narrower window starts with more knowledge, which no measurement had to pay for.

D. Van Trees bound

What we compare the strategies against is a limit on the uncertainty, whatever its source, the window or the measurements. Combining the two as independent estimates, by adding their inverse variances, gives

$$\begin{aligned} \sigma_\theta(n) &\approx \Big(J_0 + \sum_{i=1}^{n} I(x_i)\Big)^{-1/2} \\ &\geq \sigma_{\mathrm{vT}}(n) = \Big(\frac{12}{W^2} + nI^*\Big)^{-1/2}, \end{aligned} \quad (14)$$

where $\sigma_\theta(n)$ is the root-mean-square error after n measurements, averaged over the window, and the inequality uses $I(x) \leq I^*$. Eq. (14) is the van Trees inequality [28, 29], a lower bound for every estimator, biased or not, but for a Gaussian prior of variance σ_0^2. Our prior is the uniform window, so counting it as $J_0 = 12/W^2$ is a choice. It costs at most a factor 1.8 in σ, and only early in the run:

- At $n = 0$ the curve is exact: with no data the best guess is the center of the window, with error $\sigma_0 = W/\sqrt{12} \approx 0.29\, W$.
- At large n the prior term is negligible and the curve is the Cramér–Rao bound $(nI^*)^{-1/2}$ of Sec. II B, reached only by measuring every time at the steepest slope.
- In between, the hard edges of the window carry information of their own (a resonance near an edge is easier to locate), so the strict bound for a window lies below the curve, by at most $\pi/\sqrt{3} \approx 1.8$ in σ ($W/2\pi$ instead of $W/\sqrt{12}$ at $n = 0$), and meets it once the measurements outweigh the prior (Appendix C).

The bound holds for the mean squared error whatever the shape of the posterior, and for the posterior mean that error equals the expected posterior variance. The posterior standard deviation, averaged over runs, can therefore be compared with σ_{vT} from the first measurement on, as long as the posterior is calibrated. In the data limit the curve is the floor; in the crossover, a strategy may undercut it by no more than the factor 1.8.

The curve has two limits, set by how the information of the measurements compares with that of the prior. While $nI^* \ll J_0$, it stays at the prior, $\sigma_{\mathrm{vT}} \approx \sigma_0 = W/\sqrt{12}$: the measurements have not yet added to what the window told. An estimate in this limit leans on the prior and is biased toward its center, which is why its error may lie below the unbiased Cramér–Rao bound $(nI^*)^{-1/2}$. Once $nI^* \gg J_0$, the prior is washed out and $\sigma_{\mathrm{vT}} \approx (nI^*)^{-1/2}$, the unbiased Cramér–Rao bound, which falls with slope $-\frac{1}{2}$ on a log–log scale. The two limits cross at $n_c = J_0/I^*$.

The bound treats the width Γ and the contrast C as known. Inferring them as well can only add to the uncertainty in θ, but it adds nothing when the measurements are placed symmetrically on the two sides of the dip: a shift of θ then raises the signal on one side and lowers it on the other, while a change of Γ or C changes both sides alike, so the data tell the two apart.

We name the regimes after what a strategy does in them: *search* in the prior limit, where it has yet to find the dip; *refine* in the data limit, where it measures on the sides of a dip it has located; and *focus* in the crossover between them, where it narrows down from the window to the dip.

If every measurement carried I^*, the crossover would come after $n_c = J_0/I^* = \frac{64}{9}\,(\Gamma/W)^2/(k\,\mathrm{SNR}^2)$ measurements, far below one for any realistic window. It comes much later because I^* is reached only once the dip has been found: a measurement on the baseline carries $I(x) \approx 0$, so the sum in Eq. (14) stays near J_0 while the strategy searches. What such a measurement contributes instead is *elimination*: it shrinks the prior itself.

a. Elimination model. Consider an idealized adaptive strategy: it steps through the window at spacing Δx (e.g. $\Delta x = \Gamma/3$) only until it finds the dip, and from then on measures only at the steepest slope. This is not a sweep, which keeps spreading its measurements over the whole window after the dip is found (Sec. II F). Let the SNR be so high that a single measurement taken within the line, $|x - \theta| \lesssim \gamma$, shows a drop that cannot be mistaken for noise, so the first measurement that lands in the dip detects it. After n measurements without a dip, the resonance is known only to lie in the part of the window not yet excluded, and the posterior is uniform there, so

$$\sigma(n) = \frac{W - n\Delta x}{\sqrt{12}}, \qquad n < n_d, \quad (15)$$

where $n_d \leq W/\Delta x$ is the step that finds the dip (on average $W/(2\Delta x)$ over a uniformly placed resonance). In information terms $J(n) = 12/(W - n\Delta x)^2$: the gain per step is not constant but grows as the remaining width shrinks. Detection localizes the resonance to about one grid step, a uniform interval of width Δx worth $12/\Delta x^2$,

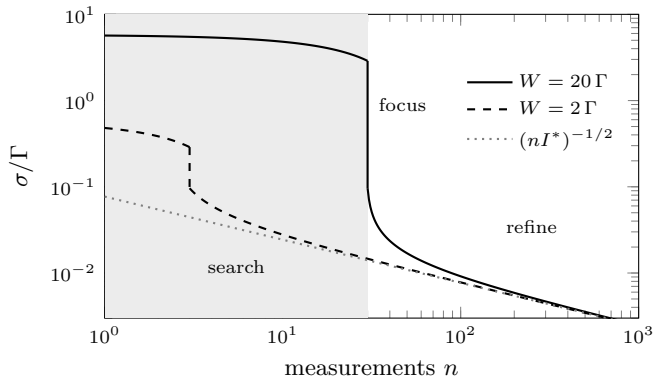


FIG. 3. Schematic of the three regimes of an adaptive strategy: the elimination model, Eqs. (15)–(16), at illustrative values (SNR 10, $k = 1$, $\Delta x = \Gamma/3$, dip found halfway through the grid). Solid: wide window, $W = 20\,\Gamma$; dashed: narrow window, $W = 2\,\Gamma$; dotted: unbiased Cramér–Rao bound $(nI^*)^{-1/2}$. The wide window shows a search plateau (shaded), a focus cliff at detection (vertical segment) and the $-\frac{1}{2}$ refine tail; the narrow window shows no plateau. Both start near the prior limit σ_0 of the van Trees bound and tend to its data limit, the unbiased Cramér–Rao bound, at large n. Simulated and measured runs are shown against the bound in Sec. IV.

after which each measurement adds I^*:

$$\sigma(n) = \Big(\frac{12}{\Delta x^2} + (n - n_d)\, I^*\Big)^{-1/2}, \qquad n \geq n_d. \tag{16}$$

Eqs. (15)–(16) reproduce both limits of Eq. (14): $\sigma = \sigma_0$ at $n = 0$ and $\sigma \to (nI^*)^{-1/2}$ for $n \gg n_d$. For a wide window the three regimes become visible (Fig. 3). The search is a plateau: Eq. (15) falls linearly, which on a log–log axis looks flat until about half the window is excluded, for a number of steps that grows with $W/\Delta x$. The focus is a cliff steeper than $-\frac{1}{2}$, where detection collapses σ from about the window to about Δx in a single step. It ends once σ is small compared with the line, so that the strategy knows where the steepest slope is. The refine regime is the $-\frac{1}{2}$ tail. A narrow window ($W \lesssim 3.5\,\Gamma$, $\sigma_0 \lesssim \Gamma$) has no plateau. The regimes belong to the adaptive strategy, which moves from the prior limit of the bound to its data limit by finding the dip first; a sweep has none, since it gains the same information $\bar{I}$ at every step (Sec. II F).

The elimination model only illustrates the shape of the profile. At low SNR one measurement no longer reveals the dip and n_d grows, but Eq. (14) holds at any SNR and needs no detection model; the runs are read against it alone.

E. Information gained per measurement

The bound counts Fisher information, the most a measurement can contribute. What a measurement actually contributes is how far it moves the posterior: the Kullback–Leibler divergence from the posterior before it to the posterior after it. Before the outcome is known, the expectation of this divergence is the expected information gain [36],

$$\begin{aligned}\mathrm{EIG}(x) &= \mathbb{E}_{y|x}\Big[D_{\mathrm{KL}}\big(p(\theta,\phi \mid y,x)\,\|\,p(\theta,\phi)\big)\Big] = I(\theta,\phi;y \mid x)\\ &= H[y \mid x] - \mathbb{E}_{\theta,\phi} H[y \mid \theta,\phi,x],\end{aligned} \tag{17}$$

the mutual information between the parameters and the outcome, with $p(\theta,\phi)$ the current posterior.

The second term is fixed by the noise model: the noise is Gaussian with the same variance σ^2/k at every drive frequency (Sec. II A), so $H[y \mid \theta,\phi,x] = \frac{1}{2}\ln(2\pi e\,\sigma^2/k)$ for every x. All the dependence on x is in the first term, the entropy of the predictive distribution of y. Its spread has two parts: the noise, and the spread of the predicted signal over the posterior,

$$\sigma^2_{\mathrm{pred}}(x) = \mathrm{Var}_{\theta,\phi}\big[\underbrace{\mu(x;\theta,\phi)}_{\mathbb{E}[y|x,\theta,\phi]}\big],$$

with θ,ϕ drawn from the current posterior. For a given variance the Gaussian has the largest entropy [37], so $H[y \mid x] \leq \frac{1}{2}\ln 2\pi e\,(\sigma^2_{\mathrm{pred}}(x) + \sigma^2/k)$ and

$$\mathrm{EIG}(x) \leq \tfrac{1}{2}\ln\Big(1 + \frac{k\,\sigma^2_{\mathrm{pred}}(x)}{\sigma^2}\Big), \tag{18}$$

with equality when $\mu(x;\theta,\phi)$, with θ,ϕ drawn from the posterior, is Gaussian distributed. Whether it is depends on the posterior of θ, not on the noise: the noise is Gaussian by construction, the posterior only becomes Gaussian with enough data.

Write $\mathrm{EIG}_n(x)$ for Eq. (17) evaluated with the posterior after n measurements. Greedy Bayesian experiment design takes the next measurement where it is largest, and summed over a run the gains are what the measurements taught, on average over runs:

$$\begin{aligned} x_{n+1} &= \arg\max_x \mathrm{EIG}_n(x),\\ \mathbb{E}\Big[\sum_{i=0}^{n-1}\mathrm{EIG}_i(x_{i+1})\Big] &= \underbrace{I(\theta,\phi;y_1,\ldots,y_n)}_{\text{learned from } n \text{ outcomes}}\,. \end{aligned} \tag{19}$$

In the refine regime Eq. (18) becomes an equality, and the EIG can be written with the Fisher information of Sec. II B. Three facts lead there.

First, the posterior of θ is a narrow Gaussian. In the data limit of Sec. II D its variance is $\sigma^2_\theta \approx 1/J_n$, one over the information collected so far, and its width σ_θ is small compared with the line.

Second, the predicted signal is then Gaussian too. Fix the drive frequency x and ask how the signal there depends on where the dip is. Moving the dip by a small amount changes the signal in proportion, at the rate $g(x)$, so over the narrow range of θ that the posterior allows, $\mu(x;\theta) \approx \mu(x;\hat\theta) + g(x)(\theta - \hat\theta)$. The predicted signal is the Gaussian θ scaled by $g(x)$: a Gaussian of width $\sigma_{\mathrm{pred}}(x) = |g(x)|\,\sigma_\theta$. Eq. (18) is then an equality.

Third, inserting this width into Eq. (18) gives $k\,g(x)^2/\sigma^2$, which is $I(x)$ by Eq. (11):

$$\begin{aligned}\mathrm{EIG}(x) &= \tfrac{1}{2}\ln\Big(1+\underbrace{\frac{k\,g(x)^2}{\sigma^2}}_{I(x)}\,\underbrace{\sigma_\theta^2}_{1/J_n}\Big)\\ &= \tfrac{1}{2}\ln\frac{\overbrace{J_n+I(x)}^{\text{after}}}{\underbrace{J_n}_{\text{before}}} = \ln\frac{\sigma_{\theta,\text{before}}}{\sigma_{\theta,\text{after}}}.\end{aligned} \tag{20}$$

The measurement adds $I(x)$ to the information J_n, so the width of the posterior shrinks from $J_n^{-1/2}$ to $(J_n + I(x))^{-1/2}$, and the EIG is the logarithm of that shrinkage. Maximizing the EIG is then maximizing $I(x)$, i.e. measuring at the steepest slope. Over a run the shrinkage factors multiply, so the gains add up to what the measurements taught, Eq. (19), and that is the logarithm of the total shrinkage:

$$\sum_n \mathrm{EIG}_n = \underbrace{I(\theta,\phi;y_1,\dots,y_n)}_{\text{learned, Eq. (19)}} = \ln\frac{\sigma_0}{\sigma_n}.$$

The information the acquisition rule maximizes and the information budget of Sec. II D are the same quantity. A strategy that measured every point at the steepest slope of the true line would reach the data limit of Eq. (14); greedy EIG approaches it once the posterior is narrow compared with the line.

Eq. (20) thus needs two conditions: the posterior has a single mode, with no appreciable mass left elsewhere in the window, and that mode is narrow compared with the line, $\sigma_\theta \ll \gamma$. Appendix E shows that the deviations from a Gaussian, and the error of the linearization, are both controlled by the ratio σ_θ/γ, and that leftover mass far from the dip dominates the posterior variance unless it is smaller than about $(\sigma_\theta/W)^2$. Both conditions are met in refine and not at the start of focus, so Eq. (20) describes the refine regime, while in focus Eq. (18) holds only as a bound. The same conditions decide when the posterior standard deviation can be read as a 68% interval: a posterior with a narrow peak and wide tails can have the same standard deviation as a Gaussian, with most of its mass much closer to the peak. [Results: when the SBED posterior becomes Gaussian, and its calibration in refine.]

In search and early focus it does not. Before the dip is found the posterior of θ is spread over the window, and the predicted signal at x is two-valued: with probability p, the posterior mass that would put the dip at x, it is about $1-C$, otherwise 1. Then $\sigma^2_{\text{pred}} = p(1-p)C^2$, while the true mutual information is at most the binary entropy $H_b(p) \le \ln 2$. The variance form keeps growing with SNR where the true gain saturates, so it overvalues search candidates relative to refine candidates; both peak at $p = \frac{1}{2}$, so they rank search candidates alike. Here greedy EIG is only myopically optimal, which is where the strategies of Sec. III B differ.

The same two-valued picture holds in focus, with a different ranking. Once a peak has formed at the dip, a small residual mass p far from it gives a predictive variance $p(1-p)C^2$ at its location, large at high SNR even for small p, while the information it holds is at most $H_b(p)$, small for small p. The variance form therefore keeps spending measurements to remove such distant mass. This reduces the posterior variance, which the tail dominates and which is the quantity Eq. (14) constrains, rather than the posterior entropy: in focus the variance rule acts as a design for the mean squared error, while a rule built on the information gain would refine the peak sooner and accept the tail. Whether this explains why focus costs SBED far more measurements than the bound is a hypothesis we have not tested. [Test: share of focus measurements placed more than a few FWHM from the peak, compared with the residual mass there.]

Bayesian SBED (Sec. III B) evaluates Eq. (18) at every candidate drive frequency on the particles of the particle filter of Sec. III A, which learns σ and uses its posterior mean. [Mention computing the exact mixture entropy from the particles, or leave out?]

F. Speedup over a drive-frequency sweep

The same budget predicts the speedup over a conventional sweep. An adaptive strategy in the refine regime measures at the steepest slope and gains, per measurement, Eq. (12),

$$I^* = \frac{27}{16}\,\frac{k\,C^2}{\sigma^2\,\Gamma^2}.$$

A sweep spreads its measurements uniformly over the window, so its information per measurement is the window average of Eq. (11),

$$\bar{I} = \frac{1}{W}\int I(x)\,dx = \frac{k}{\sigma^2 W}\int g(x)^2\,dx = \frac{\pi}{2}\,\frac{k\,C^2}{\sigma^2\,\Gamma\,W}, \tag{21}$$

using $\int g^2\,du = \pi C^2/(4\gamma) = \pi C^2/(2\Gamma)$ for the Lorentzian, valid for $W \gg \Gamma$. Both are $\propto kC^2/\sigma^2$; they differ in the width factor, $1/\Gamma^2$ against $1/(\Gamma W)$, because the sweep spends only a fraction of order Γ/W of its measurements on the line.

The sweep therefore accumulates information at the rate $\bar{I}$ from its first measurement, $\sigma \approx (J_0 + n\bar{I})^{-1/2}$, with no plateau and no cliff, as observed by Dushenko *et al.* [16]. An adaptive strategy pays the search cost n_d first and then gains I^* per measurement. Let σ_{tgt} (Hz) be the target precision, the standard deviation of f_- at which a run is considered done. The number of measurements to reach it is

$$\begin{aligned}N_{\text{sweep}}(\sigma_{\text{tgt}}) &= \frac{\sigma_{\text{tgt}}^{-2} - J_0}{\bar{I}},\\ N_{\text{ad}}(\sigma_{\text{tgt}}) &= n_d + \frac{\sigma_{\text{tgt}}^{-2} - 12/\Delta x^2}{I^*},\end{aligned} \tag{22}$$

where the adaptive count follows the elimination model: n_d measurements to detect the dip, after which the information starts at $12/\Delta x^2$ and grows by I^* per measurement until it reaches $\sigma_{\rm tgt}^{-2}$. The speedup is $R(\sigma_{\rm tgt}) = N_{\rm sweep}/N_{\rm ad}$. For a precise target the refine terms dominate and

$$R_\infty = \frac{I^*}{\bar{I}} = \frac{27}{8\pi}\frac{W}{\Gamma} \approx 1.07\,\frac{W}{\Gamma} \approx \frac{W}{\Gamma}\,: \tag{23}$$

the speedup is the inverse occupancy of the window, the number of linewidths that fit in it, which Dushenko *et al.* [16] identified qualitatively; Eq. (23) gives its prefactor. It is independent of the SNR, because both strategies gain information $\propto \mathrm{SNR}^2$. For a loose target the search cost n_d dominates $N_{\rm ad}$ and $R(\sigma_{\rm tgt}) < R_\infty$, so the full dependence on window, linewidth, SNR (through n_d at low SNR) and target is given by Eq. (22). A sweep cannot approach the van Trees bound at any n: its rate $\bar{I}$ is fixed by the window, while the bound's data limit requires every measurement at the steepest slope. [Check Eq. (23) against the speedup reported in Ref. [16] for their window and line structure (a resolved triplet, not one Lorentzian).]

G. Sensitivity and bandwidth

Sensitivity and bandwidth follow from the same budget once measurements are converted to time. Let one measurement take $t_m = k\,t_s$ (k shots of duration t_s), so that n measurements take $T = nt_m$. With D known, $D - f_- = \delta_Z = \gamma_e B_z$, so the field uncertainty is $\sigma_B = \sigma_{f_-}/\gamma_e$.

a. Sensitivity (refine). In the refine regime every measurement adds I^*, so after a time T

$$\sigma_{f_-}(T) = (nI^*)^{-1/2} = \Big(\frac{T}{t_m}\,I^*\Big)^{-1/2} \quad\Longrightarrow\quad \sigma_{f_-}\sqrt{T} = \sqrt{\frac{t_m}{I^*}}.$$

The uncertainty falls as $\sigma_{f_-} \propto T^{-1/2}$, so the product $\sigma_{f_-}\sqrt{T}$ does not depend on T and characterizes the strategy rather than the length of the run. Converted to field, this constant is the sensitivity,

$$\eta = \sigma_B\sqrt{T} = \frac{1}{\gamma_e}\sqrt{\frac{t_m}{I^*}} = \frac{1}{\gamma_e}\sqrt{\frac{16\,k\,t_s\,\Gamma^2}{27\,k\,\mathrm{SNR}^2}} = \frac{4}{3\sqrt{3}}\,\frac{\Gamma\sqrt{t_s}}{\gamma_e\,\mathrm{SNR}}, \tag{24}$$

with $\mathrm{SNR} = C/\sigma$ the single-shot SNR of Eq. (10). The number of shots k cancels as well: averaging k shots into one measurement costs k times the time and gives k times the information. The prefactor $4/(3\sqrt{3})$ is that of the standard CW-ODMR sensitivity of a Lorentzian line [32]: the familiar formula is the refine limit of the budget, reached only by a strategy that measures at the steepest slope. A strategy whose refine measurements are not all at the steepest slope gains less per measurement: if they carry on average a fraction $q \le 1$ of I^*, I^* is replaced by qI^* in Eq. (24), and its sensitivity is worse by $1/\sqrt{q}$, i.e. $\eta/\sqrt{q}$.

b. Bandwidth (whole run). A field reading is useful once its error reaches the target $\sigma_{\rm tgt}$; the time needed, $T(\sigma_{\rm tgt})$, sets the bandwidth $1/T(\sigma_{\rm tgt})$. It is t_m times the number of measurements $N_{\rm ad}(\sigma_{\rm tgt})$ of Eq. (22), n_d to find the dip and the rest to refine it to $\sigma_{\rm tgt}$:

$$T(\sigma_{\rm tgt}) = t_m\Big[n_d + \frac{\sigma_{\rm tgt}^{-2} - 12/\Delta x^2}{I^*}\Big]. \tag{25}$$

For a tight target the refine term dominates, $T \approx t_m\sigma_{\rm tgt}^{-2}/I^*$, which is the sensitivity η of Eq. (24) again. For a loose target the search term dominates, $T \approx n_d t_m$: the bandwidth cannot exceed about $1/(n_d t_m) \sim 2\Delta x/(W t_m)$, whatever the SNR. The window, i.e. the prior, therefore limits the bandwidth, while the line shape and SNR limit the sensitivity. Only the total integration time $T = nk\,t_s$ enters the bound, not how it is divided into measurements: n measurements of k shots carry the same information as nk measurements of one shot at the same drive frequency. Neglecting the overhead of switching the drive frequency, the cost of a run is therefore its total number of shots nk, the integration time in units of t_s, and strategies that group their shots differently are compared on that axis.

All closed-form results of this section are for the Lorentzian line of Eq. (6). For pulsed ODMR the same quantities follow, numerically, with Eq. (A4) in place of Eq. (6). [Which ODMR mode the results focus on, once the simulations and the experiment are complete; where pulsed goes (a Theory subsection for the information of the pulsed line, and its own Results).]

III. METHODS

A. Bayesian inference

Every strategy runs the same loop: it chooses a drive frequency x_n, takes one measurement y_n there, updates the posterior over the line parameters with it, and stops when the stop rule of Sec. III B fires. The strategies differ only in the first step; the posterior and its update, described here, are shared. The unknowns are those of Sec. II B, the parameter of interest θ and the nuisances ϕ, together with the noise level σ, which is treated separately below.

The posterior over (θ, ϕ) is a weighted cloud of particles $\{(\theta_i, \phi_i), w_i\}_{i=1}^{N_p}$, updated by sequential Monte Carlo [38]. Each particle is one hypothesis for the line. It holds the Zeeman shift $\delta_{Z,i}$, so $\theta_i = D - \delta_{Z,i}$ with D fixed (the simulations measure the mirror transition $D + \delta_{Z,i}$, which is equivalent by symmetry), and the nuisances: the linewidth itself, as the half width $\gamma = \Gamma/2$ of Eq. (6), and the contrast C. The mechanisms that set the width, Eq. (5), are not inferred; only the width they produce

is. For the pseudo-Voigt line the width is held as its Lorentzian half width and the standard deviation of its Gaussian part.

The noise is Gaussian, Eq. (8), as it is for the photon numbers of an ODMR shot, but its level σ is not known in advance: technical noise adds to the shot noise by an amount that depends on the setup and can drift. σ is therefore inferred with the line, but not as a particle dimension. Given a particle's line, the Gaussian likelihood is conjugate to an inverse-gamma prior on σ^2, so each particle carries that posterior in closed form, updated from its own residuals, and σ^2 is integrated out exactly: a Rao–Blackwellized particle filter with per-particle sufficient statistics [39, 40]. A measurement then reweights each particle by a Student-t predictive, the Gaussian likelihood averaged over that particle's noise posterior. Integrating out matters: a likelihood with a plug-in estimate of σ is largest for a particle whose σ happens to equal one residual, so resampling would favor such particles and drive σ to its lower bound, while the Student-t tails do not reward that. Old residuals are discounted, as in the variance discounting of dynamic linear models [41], so the noise posterior can follow a slow drift of σ.

When the weight has concentrated on a few particles, the cloud is resampled systematically [42] and jittered with the Liu–West kernel [43], which shrinks every particle towards the mean of the cloud and adds Gaussian noise, so that the mean and covariance of the cloud are preserved. Two safeguards depart from it: a variance floor that decays with the number of measurements, so early rejuvenations can still widen the cloud, and reflection of particles that leave the prior bounds. Each particle's noise posterior is then recomputed from the full history at its new position. The initial cloud is drawn from the prior [align with Sec. II C]. The formulas and settings are in Appendix D.

The particle filter reports the posterior mean and the weighted particle standard deviation,

$$\hat{\theta}_N = \sum_i w_i \theta_i, \qquad \hat{\sigma}_N = \Big[\sum_i w_i (\theta_i - \hat{\theta}_N)^2 \Big]^{1/2}, \quad (26)$$

the claimed uncertainty whose meaning Sec. III C discusses.

B. Measurement strategies

We compare two families of strategies. The sweep family holds the strategies an experimentalist would use without adaptive design: plain sweeps at several grid spacings, and Smart-Sweep, a sweep refined inside the dip once it is seen. The question is whether the best member of that family can match Bayesian SBED. All strategies infer with the particle filter of Sec. III A, so the comparison isolates the choice of measurements from the inference, and $\hat{\sigma}_N$ means the same for all of them. All run under the same stop rule and the same measurement budget: a run stops once the claimed uncertainty of every parameter has stayed below its ceiling for several consecutive measurements and the dip test below finds a dip.

a. Dip test. The test asks where the measurements, not the particles, show a dip. Each measurement is scored against a baseline taken from the upper part of the data, in units of the noise from the noise posterior. Contiguous stretches of measured drive frequencies, about as wide as a plausible line, get a boxcar score, and a stretch is significant when its score passes a family-wise false-alarm threshold over all stretches tested. Each connected run of significant stretches is one dip.

b. Sweep family. SimpleSweep measures a uniform grid over the window, a few points per FWHM of the prior-mean line, and repeats whole passes; the spacing is the one setting a sweep has, and several are run. Smart-Sweep sweeps the window coarsely until the dip test finds a dip, then finely inside the focus (below), and then measures only at the steepest-slope points of the posterior-mean line, once the posterior expects them to carry nearly the optimal information. Its coarse search uses the same grid points, and the same measured values, as the reference sweep of that repeat.

c. Bayesian SBED. SBED is the particle filter choosing each x by Eq. (18), maximized over candidate drive frequencies concentrated on the steep sides of the posterior-mean dip and on the dips found. The candidates are restricted to the focus: one section per detected dip, padded by one half width on each side, which widens at once when a new dip appears and narrows only after a delay. A strategy that has piled its measurements onto a suspected dip cannot be contradicted elsewhere, so with a small probability, largest while much of the window is unmeasured, SBED instead measures the middle of the widest unmeasured gap, a falsification probe.

C. Simulation setup and evaluation

a. What the claimed uncertainty is. The particle filter reports $\hat{\sigma}_N$, Eq. (26): a moment of the posterior, defined at every N. Its run average is compared with the bound from the first measurement on (Sec. II D). What fails before the posterior is Gaussian (Sec. II E) is the usual reading of $\hat{\sigma}_N$: with a uniform window or several live candidates it is not a 68% interval, and the posterior entropy is below $\frac{1}{2}\ln(2\pi e\,\hat{\sigma}_N^2)$, so information gains, per-step gains as variance ratios, and coverage at Gaussian quantiles are only meaningful in the data regime. We therefore (i) compare $\hat{\sigma}_N$ with the bound as a run average at all N, (ii) test coverage, the true error against $\hat{\sigma}_N$, only once the posterior is Gaussian, or with quantiles of the particle distribution itself, and (iii) keep this separate from miscalibration of the particle filter, which makes $\hat{\sigma}_N$ too small at any N.

b. Use cases. A use case is one simulated signal at one noise level, set by the SNR, the drive window, the Gaussian share of the width and the isotope (Ap-

pendix D). Every metric is computed within one use case, over repeats whose true signal is the same for every strategy, so strategies are compared run by run; several use cases are summarized by the median over them.

c. Metrics. The accuracy at the end of a run is not a quality metric: every run stops once its claimed uncertainty is below the ceiling, so its final error mostly reflects the ceiling. We report instead whether the claimed uncertainty is honest, through its coverage, the share of runs whose true error lies within $\hat{\sigma}_N$ and within $2\hat{\sigma}_N$; how close each strategy comes to the bound of Eq. (14), through the ratio of the bound to the mean squared true error at each N; and the cost, the number of measurements after which the true error stays below a target. Each run is also read against the bound through its information rate, the growth of $\hat{\sigma}_N^{-2}$ per measurement in units of I^*, Eq. (12); a rate above 1 means the claimed uncertainty fell faster than the data allow. Measurements to stop are compared only between strategies whose claimed uncertainty is equally honest.

D. Experimental setup

We measure the lower transition f_- only: in our setup it needs less microwave power than f_+, which keeps the heating of the stage low and the amplifier within its linear range. [Sample, optics, microwave delivery, detection and the per-shot integration time t_s.]

IV. RESULTS

A. Simulations

[(a) Claimed σ and true error against Eqs. (14)–(16) for a narrow and a wide window. (b) Information rate and prior share. (c) Cost vs SNR and window. (d) Calibration of the claimed σ.]

B. Experiment

[Placeholder until the lab data are in.]

V. DISCUSSION

[Limits: the claimed σ is overconfident; the simulated prior is miscalibrated; Gaussian noise only; I^* assumes known nuisances; low-SNR detection model.]

VI. CONCLUSION

[Conclusion.]

Appendix A: ODMR line shapes

This appendix derives the line shapes summarized in Sec. II A: the CW line with the laser treated as two rates rather than a reset, and the pulsed line with its inhomogeneous broadening and readout.

a. CW line with optical pumping. Eq. (4) treats the laser only as a reset, which is too crude when the resets are frequent. The laser acts on the spin through two rates, both set by how often the center goes through an optical cycle: it pumps the spin back toward $m_s = 0$, at a rate Γ_p, and each cycle destroys the phase between $m_s = 0$ and $m_s = -1$, at a rate Γ_c. The microwave enters only through its angular Rabi frequency $\Omega_R = 2\pi\Omega$. The competition between the two is clearest in terms of populations. A drive whose coherence is destroyed at the rate Γ_c no longer produces Rabi oscillations; it moves population between $m_s = 0$ and $m_s = -1$, in both directions alike, at the rate

$$W(\Delta) = \frac{\Omega_R^2\,\Gamma_c}{2\big[\Gamma_c^2 + (2\pi\Delta)^2\big]} = \frac{\Omega_R^2}{2\Gamma_c}\,\frac{\Gamma_c^2}{\Gamma_c^2 + (2\pi\Delta)^2}, \tag{A1}$$

a Lorentzian of peak $\Omega_R^2/2\Gamma_c$ and half width $\Gamma_c/2\pi$ in Hz. The laser returns population from $m_s = -1$ at the rate Γ_p, so with $P_0 = 1 - P_{-1}$ the population of $m_s = -1$ obeys

$$\dot{P}_{-1} = W\,(P_0 - P_{-1}) - \Gamma_p P_{-1} = W - (2W + \Gamma_p)\,P_{-1}.$$

In steady state, inserting Eq. (A1) and then ordering the terms of the denominator,

$$\begin{aligned} P_{-1}^{\mathrm{ss}}(\Delta) &= \frac{W}{2W + \Gamma_p} = \frac{1}{2}\,\frac{\Omega_R^2\,\Gamma_c}{\Omega_R^2\,\Gamma_c + \Gamma_p\big[\Gamma_c^2 + (2\pi\Delta)^2\big]} \\ &= \frac{1}{2}\,\frac{\Omega_R^2\,\Gamma_c/\Gamma_p}{(2\pi\Delta)^2 + \Gamma_c^2 + \Omega_R^2\,\Gamma_c/\Gamma_p}, \end{aligned} \tag{A2}$$

the same as the steady state of the full Bloch equations [32]. It reduces to Eq. (4) when the two rates are equal and small against Ω_R. The last form is a Lorentzian in $2\pi\Delta$ whose squared half width is the constant in its denominator, which gives Eq. (5).

The first term of Eq. (5) is the width the laser alone imposes, through the loss of coherence. The second is power broadening by the microwave: a strong drive saturates the transfer, $W \gg \Gamma_p$, already on the tails of the line, and the line widens until W there has fallen to about the pumping rate. The depth of the line, Eq. (A2) at $\Delta = 0$, is

$$P_{-1}^{\mathrm{ss}}(0) = \frac{1}{2}\,\frac{\Omega_R^2}{\Omega_R^2 + \Gamma_p\Gamma_c}\,:$$

it grows with the microwave power, toward $\frac{1}{2}$, and falls with the laser power, which repolarizes the spin before the drive can flip it.

b. Optical saturation. The two laser rates grow with the laser power, but not without limit. The optical transition is itself a two-level system: the laser excites the center at a rate proportional to its intensity $I_{\rm opt}$, and the excited state decays in its lifetime τ_e. In steady state the excited population is $s/(1+s)$, with the saturation parameter $s = I_{\rm opt}/I_{\rm sat}$ the ratio of the excitation rate to the decay rate $1/\tau_e$. At low power it grows linearly with s; at high power it levels off, because a center cannot be excited again before it has decayed. The rate of optical cycles is this population divided by τ_e, and both laser rates follow it, $\Gamma_{p,c} = \Gamma_{p,c}^{\infty}\, s/(1+s)$ [32], with $\Gamma_{p,c}^{\infty}$ their values at full saturation. Raising the laser power therefore both widens and flattens the line, up to saturation.

c. Pulsed line. The π-pulse length is chosen so that on resonance, where the spin oscillates at Ω, it is fully inverted: $\pi\Omega t_\pi = \pi/2$. Off resonance it oscillates at $\Omega_{\rm gen}$ instead, so the argument of the sine in Eq. (3) at $t = t_\pi$ is $\pi\Omega_{\rm gen} t_\pi = (\pi/2)\,\Omega_{\rm gen}/\Omega$, which gives Eq. (7). On resonance $\Omega_{\rm gen} = \Omega$ and the inversion is complete. For $|\Delta| \gg \Omega$, $\Omega_{\rm gen} \approx |\Delta|$ and $P_{-1}^{\pi} \approx \sin^2(\pi\Delta t_\pi)/(2\Delta t_\pi)^2$, the power spectrum of a rectangular pulse of length t_π: far from resonance the spin responds to the frequency content of the pulse. The line vanishes wherever $\Omega_{\rm gen} = 2\Omega, 4\Omega, \ldots$, i.e. at $|\Delta| = \sqrt{3}\,\Omega, \sqrt{15}\,\Omega, \ldots$: there the spin completes a whole number of full Rabi cycles during the pulse and returns to $m_s = 0$. Between the zeros lie side lobes that decay as $\Omega^2/\Omega_{\rm gen}^2$; the first, near $\Omega_{\rm gen} = 3\Omega$ at $|\Delta| \approx 2.7\,\Omega$, is 12% of the depth.

d. Inhomogeneous broadening. The transition frequency of the spin is not sharp: slowly fluctuating local fields, mostly from the bath of ^{13}C nuclear spins, shift it by a random offset ξ that is fixed during one pulse but changes from one repetition of the sequence to the next. A measurement averages many repetitions, each driven at the detuning $\Delta - \xi$ instead of Δ, so the measured line is Eq. (7) averaged over the distribution of ξ, i.e. convolved with it. For a Gaussian distribution of standard deviation σ_ξ,

$$\bar{P}_{-1}^{\pi}(\Delta) = \int P_{-1}^{\pi}(\Delta - \xi)\, \frac{e^{-\xi^2/2\sigma_\xi^2}}{\sqrt{2\pi}\,\sigma_\xi}\, d\xi. \qquad \text{(A3)}$$

The width σ_ξ is what T_2^* measures. A free spin precessing at the offset ξ picks up the phase $2\pi\xi t$, and the average of $e^{2\pi i\xi t}$ over the Gaussian is the coherence envelope $e^{-2\pi^2\sigma_\xi^2 t^2}$. T_2^* is defined by writing this envelope as $e^{-(t/T_2^*)^2}$, so $\sigma_\xi = 1/(\sqrt{2}\,\pi T_2^*)$. For $t_\pi \ll T_2^*$ the spread is small against Ω and the line is that of Eq. (7). For $t_\pi \gtrsim T_2^*$ the Gaussian dominates: the zeros and side lobes are washed out, the line stops narrowing at a width $\sim 1/T_2^*$, and the depth falls because the pulse no longer inverts every frequency class [32].

e. Readout. While the microwave pulse is on, the laser is off and the spin relaxes only through T_1, which is much longer than t_π, and dephases through the spread already included in Eq. (A3). The readout is different: the laser that reads the spin out also repolarizes it, so the PL difference between $m_s = 0$ and $m_s = -1$ lasts only for the repolarization time τ_p. Taking the repolarization as a single exponential, a spin flipped with probability P_{-1} emits, a time t into the readout pulse, PL at the rate $\Phi_0\,[1 - c\,P_{-1}\,e^{-t/\tau_p}]$, where Φ_0 is the rate of $m_s = 0$ and c the fractional PL drop of $m_s = -1$ at the start of the readout. The recorded PL is this rate integrated over a readout window of length t_r, normalized to the same integral with no flip,

$$\begin{aligned}\mu &= 1 - \frac{1}{t_r}\int_0^{t_r} c\,P_{-1}\,e^{-t/\tau_p}\,dt \\ &= 1 - c\,\frac{\tau_p}{t_r}\big(1 - e^{-t_r/\tau_p}\big)\,\bar{P}_{-1}^{\pi}(\Delta),\end{aligned} \qquad \text{(A4)}$$

with $P_{-1} = \bar{P}_{-1}^{\pi}$ of Eq. (A3). The integration leaves the line shape unchanged and sets only its depth: the contrast is $C = c\,(\tau_p/t_r)(1 - e^{-t_r/\tau_p})$. A longer window collects more photons, $N_{\rm ph} \propto t_r$, but dilutes the contrast, so the single-shot signal-to-noise ratio $C\sqrt{N_{\rm ph}} \propto (1 - e^{-t_r/\tau_p})/\sqrt{t_r/\tau_p}$ is largest at $t_r \approx 1.26\,\tau_p$.

Appendix B: Steepest slope of the Lorentzian line

This appendix derives the best information per measurement, Eq. (12), for the Lorentzian line of Eq. (6). Written in terms of the detuning $u = x - \theta$ of the measurement from the resonance, the line depends on x and θ only through u, so a shift of the resonance acts like the opposite shift of the measurement, and the slope of Eq. (11) is

$$g(x) = \frac{\partial\mu}{\partial\theta} = -\frac{\partial\mu}{\partial u} = -\frac{2C\gamma^2 u}{(\gamma^2 + u^2)^2}.$$

It vanishes at $u = 0$ and as $u \to \pm\infty$. Setting $\partial g/\partial u = 0$ gives $(\gamma^2+u^2)^2 = 4u^2(\gamma^2+u^2)$, i.e. $u^2 = \gamma^2/3$: the slope is steepest at $|u| = \gamma/\sqrt{3}$, where $\gamma^2/(\gamma^2+u^2) = 3/4$ and the dip is 3/4 deep. There

$$\begin{aligned} g^* \equiv \max_x |g(x)| &= \frac{2C\gamma^2\,(\gamma/\sqrt{3})}{(4\gamma^2/3)^2} = \frac{9C}{8\sqrt{3}\,\gamma} \\ &= \frac{9C}{4\sqrt{3}\,\Gamma} \approx 1.30\,\frac{C}{\Gamma}, \end{aligned}$$

with $\Gamma = 2\gamma$. Inserting g^* into Eq. (11), with $\mathrm{SNR} = C/\sigma$,

$$I^* = \frac{k\,g^{*2}}{\sigma^2} = \frac{81}{48}\,\frac{k\,C^2}{\sigma^2\Gamma^2} = \frac{27}{16}\,k\,\frac{\mathrm{SNR}^2}{\Gamma^2}.$$

Appendix C: The uniform window and the van Trees bound

This appendix derives the statements of Sec. II D about counting the uniform window as $J_0 = 12/W^2$.

a. The prior's own information. For a prior density $p(\theta)$, the van Trees inequality [28, 29] bounds the mean squared error of any estimator, averaged over the prior, by

$$\sigma_\theta^2(n) \;\geq\; \Big(J(p)+\sum_{i=1}^{n} I(x_i)\Big)^{-1}, \qquad J(p)=\int \frac{p'(\theta)^2}{p(\theta)}\,d\theta.$$

$J(p)$ measures how sharply the prior changes when θ moves. Inside a uniform window it does not change at all, but at the two edges it drops from $1/W$ to zero in a single step, so $J(p)$ is infinite and the inequality reduces to $\sigma_\theta \geq 0$: true, but empty. The edge stands for infinitely sharp knowledge, that θ is certainly not even slightly beyond it. Behind this there is a real but finite effect: a resonance close to an edge is easier to locate, because the posterior is cut off at the edge. The same holds for any prior with hard edges, whatever part of the scanned range it covers; only a smooth prior has a finite $J(p)$.

b. Why the inverse variance. For any prior, $J(p) \geq 1/\mathrm{Var}(\theta)$, the Cramér–Rao inequality applied to a shift of the prior itself [35], with equality only for a Gaussian. With $J_0 = 1/\sigma_0^2$ in place of $J(p)$, Eq. (14) is therefore the van Trees bound for a Gaussian prior of the same variance, and for any other prior it lies at or above the strict bound: it is not itself a lower bound. Where the signal is linear in θ over the posterior, as in the data limit, the best estimate that is linear in the data has error $(J_0+\sum_i I(x_i))^{-1/2}$ for any prior of variance σ_0^2, because it depends on the prior only through its mean and variance [28]. The best estimate overall can only do better, so there the true optimum lies at or below the curve.

c. A strict bound on the window. Among all densities confined to the window, the one with the least $J(p)$ is $p(v) = (2/W)\cos^2(\pi v/W)$, with v the distance from the center, and $J(p) = 4\pi^2/W^2 \approx 3.3\,J_0$ [29]. The van Trees inequality with this prior bounds the error averaged over that prior and hence the worst-case error over the window, but not the error averaged over the uniform prior. It lies below Eq. (14) by a factor of up to $\pi/\sqrt{3} \approx 1.8$ in σ, all of it in the prior limit and the crossover: at $n = 0$ it gives $W/2\pi \approx 0.16\,W$, against the exact $W/\sqrt{12} \approx 0.29\,W$, and in the data limit the prior term is negligible and the two coincide. Which prior enters the bound therefore matters only before the measurements dominate, in the prior limit and the crossover, and not in the data limit.

Appendix D: Algorithm settings

a. Particle filter. The cloud has $N_p = 1000$ particles. Every dimension is scaled to $[0,1]$ by its prior bounds, so one resampling kernel serves all of them; reported values are mapped back to Hz. The cloud is resampled when the effective sample size $(\sum_i w_i^2)^{-1}$ falls below $0.2\,N_p$, systematically, and then rejuvenated with the Liu–West kernel,

$$\psi_i' \sim \mathcal{N}\big(a\psi_i + (1-a)\bar{\psi},\; (1-a^2)\Sigma\big), \qquad a = 0.98, \quad \text{(D1)}$$

with $\psi_i = (\theta_i, \phi_i)$ the whole particle and $\bar{\psi}$, Σ the weighted mean and covariance before resampling. The diagonal of Σ is floored at $(0.01\,e^{-n/25})^2$ of each unit range after n measurements, and a particle that leaves the unit cube is reflected back into it.

b. Noise model. Each particle carries an inverse-gamma posterior on σ^2 with shape α, the same for every particle because its update does not depend on the residuals, and scale β_i. The prior is $\alpha_0 = 10$, $\beta_0 = \alpha_0\sigma_{\mathrm{nom}}^2$, with σ_{nom} the geometric mean of the prior range of σ. Before each measurement both are discounted by $d = 0.99$; a measurement y_n, the mean of k shots at x_n, then adds $\frac{1}{2}$ to α and $\frac{1}{2}k\,r_{n,i}^2$ to β_i, with residual $r_{n,i} = y_n - \mu(x_n;\theta_i,\phi_i)$, so that after n measurements $\beta_i = d^n\beta_0 + \frac{1}{2}\sum_{t<n} d^{\,n-1-t} k\, r_{t,i}^2$. Old residuals are thus forgotten with a memory of about $1/(1-d) = 100$ measurements, and α saturates at $0.5/(1-d) = 50$, about 100 degrees of freedom. With σ^2 integrated out, the measurement reweights the particles by the Student-t predictive

$$w_i \;\propto\; w_i\; t_{2\alpha}\Big(y_n;\, \mu(x_n;\theta_i,\phi_i),\, \frac{\beta_i}{k\alpha}\Big), \qquad \text{(D2)}$$

taken with α, β_i before the update. After a rejuvenation each β_i is recomputed from the full history at the particle's new position.

c. Initial cloud. [The prior on the Zeeman shift δ_Z is being changed to uniform over the range the truth is drawn from (now a truncated Gaussian, offset from the truth, plus 10% of the particles uniform over the bounds); state it here once Sec. II C is fixed.]

d. Stop rule and budget. The ceilings on the claimed uncertainty are 100 kHz for θ, 500 kHz for each width and 0.01 for the contrast; they must hold for 8 consecutive measurements, together with a dip found by the dip test, which is re-run at most every 10 measurements. Every strategy has the same budget of 2500 measurements; a sweep's total is capped in whole passes. A run that reaches the budget without stopping is censored and counted at the budget, as a lower bound on its cost.

e. Dip test. Each measurement gets the score $z_n = (b - y_n)/s$. The baseline b is the 70th percentile of the y_n minus its expected offset under pure noise, $t_\nu^{-1}(0.7)\,s$, and s and ν are the scale and degrees of freedom of the noise posterior. Every contiguous stretch of measured drive frequencies whose span lies in the prior range of the linewidth gets the boxcar score $\sum z_n/\sqrt{n}$, over its n measurements. A stretch is significant when its score exceeds $t_\nu^{-1}(1 - p_{\mathrm{FA}}/N_{\mathrm{tests}})$, the Bonferroni threshold for a family-wise false-alarm probability $p_{\mathrm{FA}} = 0.1$ over the N_{tests} stretches tested.

f. Sweep family. Γ_0 is the FWHM of the prior-mean line. SimpleSweep measures a uniform grid of spacing Γ_0/p, $p = 2, 3, 4$. Smart-Sweep sweeps at $\Gamma_0/3$ until the

dip test finds a dip, then at $\Gamma_0/10$ inside the focus, and then measures only at the maxima of $g(x)^2$ at the posterior mean, cycling between them, once the posterior expects each of those points to carry at least 90% of the information of the true optimum.

g. Bayesian SBED. After every resampling SBED places 800 candidate drive frequencies at the quantiles of a density with Gaussian kernels on the two sides of the posterior-mean dip, ±HWHM from its center, kernels on the dips found and a flat share, thinned to a spacing of at least 100 kHz. Eq. (18) is evaluated on a 500-particle subsample, with σ^2 the posterior mean of the noise. The candidates are taken in blocks of 64, and the block maxima compete through a softmax with a temperature of 0.01 nats, so the choice is near-greedy. Each focus section is a dip's extent padded by one posterior-mean HWHM on each side. A new dip widens the focus at once; narrowing waits 8 measurements and must shrink the focus by at least 5%. The focus is re-decided after every resampling. The falsification probe is made with probability $0.02 + 0.08\,g$, where g is the widest unmeasured gap as a fraction of the window.

h. Use cases. The simulated signals are pseudo-Voigt dips of total FWHM 2 MHz, on a grid of the SNR, the window (1.6, 2, 3 and 5 times the width one dip group covers), the Gaussian share of the width (0, 0.15, 0.3) and the isotope (unresolved hyperfine lines, ^{14}N or ^{15}N). The code labels the SNR levels 1, 2, 5, 10, 20 against a contrast of 0.25 [convert to C/σ of Eq. (10): the code's contrast is $2C$]. [Number of repeats per use case.]

Appendix E: Approach of the posterior to a Gaussian

Eq. (20) and the reading of σ_θ as a 68% interval assume a Gaussian posterior. This appendix states when that holds for the line of Eq. (6). There are two separate questions: whether the posterior has a single peak, and whether that peak is Gaussian. We take the nuisances as known; they add dimensions but do not change the scaling.

a. Shape of a single peak. Let $L(\theta) = \ln p(\theta) + \sum_i \ln p(y_i \mid \theta, x_i)$ be the log posterior and $\hat\theta$ its maximum. Inside the window the uniform prior is flat and only the likelihood contributes. Expanding about the maximum (Laplace's method [44]),

$$L(\theta) = L(\hat\theta) - \frac{J}{2}(\theta-\hat\theta)^2 + \frac{L'''}{6}(\theta-\hat\theta)^3 + \frac{L''''}{24}(\theta-\hat\theta)^4 + \dots, \tag{E1}$$

with $J = -L''(\hat\theta)$, whose expectation is the Fisher information J_n of Eq. (14). Keeping only the quadratic term gives a Gaussian of variance $\sigma_\theta^2 = 1/J$. Over the posterior, $|\theta - \hat\theta| \sim \sigma_\theta$, so the higher terms change the log density by relative amounts of order $L'''\sigma_\theta^3$ and $L''''\sigma_\theta^4$; the first sets the asymmetry of the peak, the second the weight of its tails.

For Gaussian noise, $\ln p(y \mid \theta, x) = -k\,[y - \mu(x;\theta)]^2/2\sigma^2$ up to a constant. Averaged over the noise, the derivatives are

$$\mathbb{E}[L''] = -\sum_i \frac{k\,\mu_i'^2}{\sigma^2} = -J_n, \qquad \mathbb{E}[L'''] = -3\sum_i \frac{k\,\mu_i'\mu_i''}{\sigma^2},$$

where primes are derivatives with respect to θ at x_i, so $\mu_i' = g(x_i)$. The line changes on the scale of its half width γ, so each further derivative brings a factor of order $1/\gamma$: $\mu'' \sim \mu'/\gamma$, hence $|L'''| \lesssim J_n/\gamma$ and $|L''''| \lesssim J_n/\gamma^2$. The corrections in Eq. (E1) are then of relative order

$$L'''\sigma_\theta^3 \sim \frac{\sigma_\theta}{\gamma}, \qquad L''''\sigma_\theta^4 \sim \frac{\sigma_\theta^2}{\gamma^2}. \tag{E2}$$

A single peak is Gaussian once it is narrow compared with the line, $\sigma_\theta \ll \gamma$. This is an upper estimate: at the steepest-slope points $\mu'' = 0$ (there $|g|$ is at its maximum), so for measurements placed there the cubic term vanishes on average and the corrections start at order σ_θ^2/γ^2.

In the data limit, $J_n \approx nI^*$ with I^* of Eq. (12), and with $\Gamma = 2\gamma$,

$$\frac{\sigma_\theta}{\gamma} \approx \frac{8}{3\sqrt{3}}\,\frac{1}{\mathrm{SNR}\sqrt{nk}} \approx \frac{1.5}{\mathrm{SNR}\sqrt{nk}},$$

so at an SNR of order 10 a few shots on the sides of the dip already make the peak Gaussian.

b. Linearization of the signal. The same ratio controls the step from a Gaussian posterior to a Gaussian prediction used in Eq. (20). Expanding the signal over the posterior, $\mu(x;\theta) = \mu(x;\hat\theta) + g(x)(\theta-\hat\theta) + \frac{1}{2}\mu''(x)(\theta-\hat\theta)^2 + \dots$, the quadratic term relative to the linear one is $\mu''\sigma_\theta/2g \sim \sigma_\theta/\gamma$. Eq. (20) therefore holds up to relative corrections of order σ_θ/γ, and of order σ_θ^2/γ^2 at the steepest slope.

c. Leftover mass. The expansion describes one peak. Early in focus the posterior is instead a mixture: a peak of weight $1-p$ and width σ_θ at the dip, and a remainder of weight p spread over the part of the window not yet excluded, of width up to W. The variance of the mixture is

$$\mathrm{Var}(\theta) \approx (1-p)\,\sigma_\theta^2 + p\,s_\mathrm{r}^2,$$

where $s_\mathrm{r}^2 \lesssim W^2/12$ is the mean squared distance of the remainder from the peak. The remainder dominates once $p \gtrsim \sigma_\theta^2/s_\mathrm{r}^2$, i.e. roughly $p \gtrsim 12(\sigma_\theta/W)^2$, a very small weight: for $W = 20\,\Gamma$ and $\sigma_\theta = 0.1\,\Gamma$ it is 3×10^{-4}. Until the remainder has fallen below this, the posterior standard deviation reports the remainder, not the error of the peak, and the posterior is far from Gaussian: a narrow peak with wide tails. Its entropy, by contrast, is changed by the remainder only by about $H_b(p) + p\ln(s_\mathrm{r}/\sigma_\theta)$, which is small for small p. This is the difference between the variance and the information gain discussed in Sec. II E.

The remainder is removed only by measuring where it would place the dip. A measurement within about

γ of such a location, when the dip is not there, lowers the log weight of that part of the remainder on average by the Kullback–Leibler divergence between the two predicted outcomes, $kC^2/2\sigma^2 = k\,\mathrm{SNR}^2/2$. Lowering a weight p to below $12(\sigma_\theta/W)^2$ takes, at each location, about $2\ln[pW^2/12\sigma_\theta^2]/(k\,\mathrm{SNR}^2)$ measurements, and at least one; a remainder spread over the whole window has of order W/Γ such locations. At high SNR the posterior therefore becomes Gaussian after about one pass over the remaining window, a cost that grows with W/Γ like that of the search. [Compare with when the SBED posterior becomes Gaussian in the runs (Results).]

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